CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-015

WASTE DISCHARGE REQUIREMENTS
FOR
DIXON MIGRANT CENTER

CITY OF DIXON HOUSING AUTHORITY
YOLO COUNTY HOUSING AUTHORITY
CALIFORNIA DEPARTMENT OF HOUSING AND COMMUNITY DEVELOPMENT,
UNITED STATES DEPARTMENT OF THE NAVY, AND
JOHN AHMANN
SOLANO COUNTY

The California Regional Water Quality Control Board, Central Valley Region (hereafter Board), finds that:

1. On 8 November 1999, a Report of Waste Discharge (RWD) was submitted by the Yolo County Housing Authority for the Dixon Migrant Center to include alternative discharge options for the domestic wastewater. The RWD was found to be incomplete and on 9 June 2000, an amendment was submitted to complete the RWD.

2. Waste Discharge Requirements (WDRs) Order No. 95-128, adopted by the Board on 26 May 1995, prescribes requirements for the discharge of domestic wastewater to evaporation/ percolation ponds. Proposed expansion of the domestic wastewater disposal facilities and off-site recycling of the domestic wastewater requires that the WDRs be revised. In addition, Order No. 95-128 is neither adequate nor consistent with current plans and policies of the Board.

3. The Dixon Migrant Center, aka Rehrman Migrant Center, is on the property currently owned by the United States Department of the Navy (Navy), and is known as the Naval Radio Transmitting Facility (NRTF), Dixon. The Navy is the land owner/administrator. The migrant center and the wastewater pond properties are in the process of being declared surplus by the federal government. The United States Government Services Administration (GSA) will administer this process.

4. The Dixon Migrant Center is currently operated by the Yolo County Housing Authority, through a Joint Powers Agreement with the City of Dixon Housing Authority. The City of Dixon Housing Authority operates under a yearly contract from the California Department of Housing and Community Development, Office of Migrant Services (Department). Historically, the City of Dixon Housing Authority operated the migrant center, but in March 1997, the Department requested that the Yolo County Housing Authority be contracted by the City of Dixon Housing Authority for operation of the center.

5. The Dixon Housing Authority, Yolo County Housing Authority, the Department (operators), the Navy and John Ahmann (owners) are hereafter Discharger.

6. The Dixon Migrant Center has been in operation since 1984 and consists of separate housing and wastewater disposal areas. Both areas are at the Naval Radio Transmitting Station, 7290 Radio Station Road, approximately 6 miles southeast of Dixon in Solano County. The housing area occupies approximately 27 acres (Assessor's Parcel No 0143-060-070) and the domestic wastewater ponds occupy approximately 5-acres of a 33-acre parcel (Assessor's Parcel No 0143-060-110). The migrant center is in the northwest quarter of the northeast quarter, and the domestic wastewater ponds are in the eastern half of the southeast quarter of Section 8, T6N, R2E, MDB&M, as shown on Attachments A and B and part of this Order by reference.

7. Domestic wastewater from the Navy operations at the NRTF, Dixon is not discharged to the two ponds operated for the migrant center. A separate system is used to treat and dispose of the wastewater, which consists of a septic tank and leachfield followed in series by a pond.

8. The Migrant Center has 20 single family dwellings and 24 duplexes to house residents, a community center, day-care center, laundry facility, office, storage area, and maintenance shop; all fully equipped with domestic wastewater services. The site plan is shown on Attachment C, which is attached hereto and is part of this Order by reference.

9. The Dixon Migrant Center provides housing for migrant workers and their families during the harvest season, generally occurring from 1 May through 31 October (i.e. harvest season). There are approximately 370 residents and 15 teachers at the migrant center, when fully occupied. In the off-season, approximately 15 people (3 maintenance workers and their families) live at the center.

10. The domestic wastewater system consists of a 32, 000 gallon septic tank, a pump station equipped with two 5-horsepower (HP), alternating cycle submersible pumps, and two oxidation ponds. Wastewater is pumped through approximately 3,350 feet of 4-inch polyvinyl chloride (PVC) pipe to the ponds. The pond plan and specifications are shown on Attachment D, which is attached hereto and is part of this Order by reference. A flow meter was installed in the force main in January 2000.

11. The two ponds, known as Pond 1 (southern) and Pond 2 (northern), were constructed in 1984 to replace smaller wastewater ponds, and are shown on Attachment D, which is attached hereto and is part of this Order by reference. Ponds 1 and 2 are adjacent to the eastern edge of the Navy property, which is bounded by an agricultural ditch. Pond 1 is 285 feet by 385 feet and Pond 2 is approximately 425 feet by 220 feet. Both ponds are approximately 5 feet deep with design water depths of 2.55 and 2.25 feet for Ponds 1 and 2, respectively.

12. Wastewater can flow into either pond, but is currently valved to flow into Pond 2, then to Pond 1, through a transfer structure pipeline in the common levee. A staff gauge has been installed near

the southeast corner of each pond. The design capacity of Ponds 1 and 2, with 2 feet of freeboard, is 3.75 and 3.72 acre-feet, respectively, due to the slope of the levees.

13. The RWD indicated that average daily wastewater flows during the harvest season are estimated to be approximately 23,000 gallons per day (gpd). During the off-season, average daily wastewater flows are estimated to be less than 4,000 gpd. Design flows for the harvest and off-seasons are estimated to be approximately 25,000 gpd and 6,000 gpd, respectively. However, recent meter readings (from September 2000) indicate that flows are as high as approximately 37,000 gpd, with a daily average flow of approximately 25,000 gpd.

14. The RWD estimates that during the 6-month season of occupancy, the average biochemical oxygen demand (BOD) loading rate in the ponds is 19 pounds per acre per day (lbs/ac/day), as calculated based on 0.17 lbs/person/day.

15. WDRs Order No. 95-128 does not require water quality monitoring. Therefore, an analytical characterization of the pond wastewater has not been conducted. However, field data, collected in November 2000 had the following results:

| Pond | pH[1] | Electrical Conductivity (EC)[2] |
|---|---|---|
| 1 | 8.1 | 1890 |
| 2 | 9.2 | 1502 |

pH = pH Units
EC = micromhos per centimeter (μmhos/cm)

16. Since adoption of WDRs Order No. 95-128, the migrant center's wastewater ponds have experience operational problems. Insufficient storage capacity of the ponds and the threatened discharge to surface water drainage courses were identified during inspections conducted by Regional Board staff on 8 May 1996 and 18 June 1999, and during the 9 September 1996 and 10 February 1999 inspections conducted by the Navy. During these inspections, the presence of a bypass valves from each of the ponds and a discharge pipe into the agricultural ditch were noted.

17. In September 1999, the bypass valves were plugged with cement and the discharge point to the agricultural ditch was removed. Beginning in 1997 and continuing in 1999 to 2000, several improvements were made to the domestic wastewater system. Major improvements included cleaning and repair of the septic tank; modification to sewer manholes to eliminate surface water inflow; installation of caps on overflow, seasonal, and emergency pond drain lines; replacement of sewer collection pipelines and force main; repair of the pond distribution boxes; initiation of a weed control program; and employment of a certified operator.

18. Proposed improvements to the wastewater ponds include minor grading of the levee crowns to eliminate puddling of rainwater, and installation of riprap on the interior levee slopes to restore embankment integrity.

19. The RWD states that studies of flow meter data from January through April 2000 indicate that there is a significant infiltration from groundwater into the collection system. Based on these preliminary measurements, infiltration has been estimated as contributing up to 5,000 gpd of the daily flows.

20. During a meeting in January 2000, potential discharge alternatives were discussed to address the apparent problems with the wastewater pond capacity. Short-term alternatives included off-site discharge, and discharge to a 15-acre parcel, immediately north of the wastewater ponds, was considered a long-term discharge alternative. The Navy currently owns the 15-acre parcel.

21. The RWD proposed backup disposal capacity for the domestic wastewater based on the findings of the hydraulic balance for the 100-year annual return. Backup capacity may be used during maintenance activities or during an emergency situation that could occur with excessive rainfall. Excess wastewater during these situations would be either recycled off-site to the property immediately east of the ponds, or would be discharged to the 15-acre parcel.

22. The off-site property (633-acre parcel), which is designated for wastewater recycling, is to the east of the domestic wastewater ponds, and is in Section 9, T6N, R2E, MDB&M, as shown on Attachment B. The property is owned by John Ahmann (Assessor's Parcel No 143-070-010). This parcel is currently used as non-irrigated cattle pasture. Flood irrigation is proposed to enhance the growth of native grasses. Wastewater is proposed to be transferred to the off-site property using a portable, engine-driven pump and plastic piping.

23. A written lease agreement between John Ahmann and the Yolo County Housing Authority was executed, with annual renewals, to use the adjacent land to recycle the wastewater. The final signature date on the lease agreement is 27 July 2000.

24. A Title 22 Engineering Report (Report) was submitted on 21 June 2000 for the off-site recycle alternative. The California Department of Health Services approved the Report on 8 August 2000.

25. Prior to execution of the legal agreement for the off-site recycling of the wastewater, a composite soil sample was collected from the Ahmann property. The analytical results of the soil sample is as follows:

| Compound | Units | Concentration |
|---|---|---|
| pH | pH Units | 6.6 |
| Nitrate (as Nitrogen) | mg/kg | 1 |
| Phosphorus | mg/kg | 8 |
| Potassium | mg/kg | 270 |
| Sodium | mg/kg | 190 |
| Copper | mg/kg | 4.5 |
| Iron | mg/kg | 47 |
| Zinc | mg/kg | 0.7 |
| Soluble Salts | mmhos/cm | 0.2 |
| Sodium Adsorption Ratio | ratio | 1.2 |

mg/kg = milligrams per kilogram
mmhos/cm = millimhos per centimeter

26. Use of the 15-acre parcel is also being considered as a secondary backup capacity alternative for disposal of the domestic wastewater. The Navy has identified the parcel as surplus and is proposing to include it with the transfer of the migrant center and wastewater pond parcels. The transfer of ownership, once approved by the GSA, is anticipated to be complete in approximately three years. Should this property transfer occur, the RWD proposes the wastewater be discharged either to a pond or land application area system.

27. Potable water is supplied from two wells (Well Nos. 1 and 2) that are operated by the Navy. Well No. 1 is immediately northeast of the water storage tanks in the southern portion of the site, and Well No. 2 is along the southern fence line of the migrant center. Construction information for Well No. 1 is not available. However, the driller's log for Well No. 2 indicates that the borehole was drilled to 241 feet below ground surface (bgs) and the well was installed to 230 feet bgs, with a screen interval from 128 to 136 feet bgs and 195 to 226 bgs. Both wells are reported to have 5-horsepower submersible pumps that supply water at approximately 125 gallons per minute.

28. Drinking water quality from the supply wells is monitored in accordance with the requirements of the California Department of Health Services, San Francisco District. Field data from Well No. 1, collected in November 2000, had a pH of 8.74 and 595 μmhos/cm for EC.

29. The Navy has entered into negotiations with the Department and Yolo County Housing Authority for installation of a new supply well for the migrant center. The Navy requested that the new well be installed prior to transfer of the property. Engineering studies for a new water supply system are now being conducted through the Yolo County Housing Authority.

30. There is no information on shallow groundwater for this site, as monitoring wells have not been installed. Depth to groundwater has not been quantified, but is relatively shallow year-round, as

it was estimated in the RWD to be approximately 4 feet below ground surface (bgs) from February to April 2000.

31. Based on observations during improvements to the domestic wastewater system, soil in the upper few feet of the subsurface has been reported to consist of clay to silty-clay. The *Soil Survey of Solano County* (U. S. Soil Conservation Service) classifies the soil at the migrant center as Capay Clay, and the soil in the area of the wastewater ponds as part of the Antioch-San Ysidro complex and the Capay Clay. Based on the driller's log from supply Well No. 2, soil in the upper 15 feet of the soil horizon is primarily described as a clay, with stringers of sandy clay to a depth of 132 feet bgs.

32. Local land use is for radio transmitting and receiving antenna operated by the Navy. The Navy also leases large portions of the NRTF, Dixon for farming. Sugar beets are cropped to the west and south of the wastewater ponds. The 15-acre parcel immediately north of the ponds, slated for disposal by the Navy, is currently cropped with alfalfa.

33. The ground surface elevation near the Discharger's wastewater ponds ranges from approximately 23.5 to 25 feet mean sea level (msl). The pond levees have an average elevation of 26.55 feet msl. The approximate elevation of the bottom of the ponds is 22 feet msl. The ground surface elevation on the Ahmann property is approximately 24 feet msl with surface drainage to the southeast.

34. As reported in the RWD, precipitation ranges from no measurable rainfall in July/August to a maximum of 4.16 inches in January. The mean annual rainfall is 16.83 inches, as measured at the Davis station (Department of Water Resources, Division of Flood Management). The RWD estimates the evaporation rate as ranging from a minimum of 1 inch in January to a maximum of 7.8 inches in July.

35. Storm water runoff sheet drains to the southeast following the topography and does not drain through the domestic wastewater collection system.

36. Surface water drainage in the domestic wastewater ponds is to an agricultural ditch along the eastern border of the NRTF, Dixon. Surface water then drains to a lateral managed by Reclamation District No. 2068 (RD 2068). Surface water runoff from the Ahmann property drains to RD 2068's laterals along the southern and eastern boundaries of the section. RD 2068's laterals drain to Haas Slough, tributary to Cache Slough, tributary to the Sacramento River. The facility is in the Elmira Hydrologic Subarea (No. 511.10), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

37. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento/San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

38. The beneficial uses of the Haas Slough, Cache Slough and the Sacramento River are municipal and domestic supply, agricultural supply for irrigation and stock watering, contact and non-contact recreation, warm freshwater habitat, fish spawning, wildlife habitat, and groundwater recharge.

39. The beneficial uses of ground water are domestic, industrial, process and agricultural supply.

40. Water quality criteria associated with the protection of groundwater beneficial uses will be determined during the Well Installation and Groundwater Characterization Report required by this Order.

41. The Board has considered antidegradation pursuant to State Board Resolution No. 68-16 and finds that the discharge is consistent with those provisions, and should not cause an increase in groundwater constituents above that of background levels. However, additional data will be required to substantiate this finding.

42. The action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with California Code of Regulations (CCR), Title 14, Section No. 15261. This exemption reflects the ongoing nature of the operator's wastewater facilities project, originally approved in 1983, and does not provide for any net increase in treatment or disposal capacity from WDRs Order No. 95-128.

43. This discharge is exempt from the requirements of Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereinafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:

    a. The Board is issuing waste discharge requirements, and
    b. The discharge complies with the Basin Plan, and
    c. The wastewater does not need to be managed according to 22 CCR, Division 4.5, Chapter 11, as a hazardous waste.

44. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could

affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."

45. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

46. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

**IT IS HEREBY ORDERED** that Order No. 95-128 is rescinded and the Dixon Housing Authority, Yolo County Housing Authority, the California Department of Housing and Community Development, the U.S. Department of the Navy, and John Ahmann, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

**A. Discharge Prohibitions:**

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited, except as discussed in the Standard Provisions.

3. Discharge of waste classified as 'hazardous,' defined in Section 20164 of Title 27, CCR, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.

4. The discharge of wastewater from the ponds to the recycling area and/or land application area is prohibited during the period extending from 1 November to 15 April.

5. The discharge of irrigation tail waters beyond the boundaries of the Ahmann property, or beyond the boundaries of the 15-acre parcel, which are designated for wastewater recycling area and land application area, respectively, is prohibited.

6. The discharge of solvents, degreasers, or any other cleaners, other than biodegradable soaps to the wastewater treatment and disposal system, is prohibited.

**B. Discharge Specifications:**

1. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined in the California Water Code, Section 13050.

2. The discharge shall not cause degradation of any water supply.

3. The septic tank shall have the necessary capacity to provide adequate detention time to remove scum and settleable solids from the wastewater flow.

4. The treatment and disposal facilities shall be operated and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

5. The monthly average dry weather flow from the migrant center shall not exceed 25,000 gpd. The daily maximum flow from the migrant center shall not exceed 37,000 gpd.

6. The wastewater ponds shall have sufficient capacity to accommodate allowable wastewater, and design seasonal precipitation and ancillary inflow and infiltration, in order to prevent inundation or washout due to floods with a 100-year return frequency.

7. A minimum freeboard of 2.0 feet (measured vertically to the lowest point of the berm) shall be maintained in the ponds at all times.

8. The discharge of wastewaters from the migrant center shall be to the wastewater ponds at all times.

9. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the property owned by the Discharger.

10. As a means of discerning compliance with Discharge Specification No. 9, the dissolved oxygen content in the upper zone (1 foot) of the wastewater ponds shall not be less than 1.0 milligram per liter (mg/l).

11. The wastewater ponds shall not have an average daily pH of less than 6.5 or greater than 9.0.

12. The wastewater ponds shall be managed to prevent breeding of mosquitoes. In particular,

    a. An erosion control program shall assure that small coves and irregularities are not created around the perimeter of the water surface.
    b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

13. Public contact with wastewater in the wastewater ponds shall be precluded through such means as fencing, signs and other acceptable alternatives. All signs shall be posted in English and Spanish.

**C. Recycled Water Specifications (633-acre Ahmann property):**

1. The discharge of recycled water from the wastewater ponds shall be to the Ahmann property only.

2. Recycled water shall remain within the recycling area at all times.

3. Recycled water used for irrigation shall be managed to minimize erosion.

4. The Discharger shall maintain the following minimum setback distances from areas irrigated with recycling water:

| Minimum Setback Distance | To |
|---|---|
| 25 | Property Line |
| 30 | Public Roads |
| 50 | Drainage Courses |
| 150 | Domestic/Irrigation wells |

5. Application of the recycled water to the recycling area shall be managed to minimize percolation to groundwater.

6. There shall be no visible standing water in the recycling area for more than 48 hours in any consecutive 7-day period after application of the wastewater.

7. Application of recycled water to the recycling area shall be at reasonable rates considering the crops or vegetation, soil, climate, and irrigation management system.

8. Recycled water shall be applied to the designated recycling area at agronomic rates and shall not exceed the average monthly evapotranspiration of the recycling area.

9. Areas irrigated with recycled water shall be managed to prevent breeding of mosquitoes, where:

a. Ditches are maintained to be free of emergent, marginal, and floating vegetation; and

b. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store water.

10. All areas where recycled water is to be used shall be posted with conspicuous signs that present the following wording that can be read by the public in English and Spanish:

"RECYCLED WATER – DO NOT DRINK – WASH THOROUGHLY WITH SOAP AND DRINKING WATER IF CONTACT OCCURS."

11. Recycled water shall be managed to minimize contact with workers.

12. The user shall not irrigate with recycled water during periods of heavy rainfall or when the ground is saturated.

**D. Land Application Area Specifications (15-acre Navy property):**

1. Treated wastewater shall remain within the land application area at all times.

2. Treated wastewater used for evaporation/percolation shall be managed to minimize erosion.

3. Application of the treated wastewater to the land application area shall be managed to minimize percolation to groundwater.

4. There shall be no visible standing water in the land application area for more than 48 hours in any consecutive 7-day period after application of the wastewater.

5. Areas discharged in the land application area shall be managed to prevent breeding of mosquitoes, where:

   a. Ditches are maintained to be free of emergent, marginal, and floating vegetation; and
   b. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store water.

6. The user shall not discharge to the land application area during periods of rainfall or when the ground is saturated.

7. Public contact with wastewater in the land application area shall be precluded through such means as fencing, signs and other acceptable alternatives. All signs shall be posted in English and Spanish.

**E. Sludge Disposal:**

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 27, Division 2, Subdivision 1 of the CCR and approved by the Executive Officer.

2. Use and disposal of sewage shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in the Code of Federal Regulations (CFR), 40 CFR 503.

   If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

3. Any proposed change in sludge disposal from a previously approved practice (as described in these WDRs) shall be reported to this office at least 90 days in advance of the change.

4. The septic tank shall be inspected to assure adequate treatment and biosolids removal. Septic tanks shall be pumped annually or when any one of the following conditions exists or is projected to occur before the next inspection:

   a. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment; or
   b. The scum layer is within three inches of the outlet device; or
   c. The sludge layer is within eight inches of the outlet device.

**F. Ground Water Limitations:**

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed a median value of 2.2/100 ml over any 7- day period.

**G. Provisions:**

1. The Discharger shall comply with the Monitoring and Reporting Program (MRP) No. 5-01-015, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. The Discharger shall notify Reclamation District No. 2068 in the event of a breach of the wastewater ponds or with annual commencement of wastewater recycling. Notification of breach conditions shall be provided upon discovery; notification of wastewater recycling shall be provided seven calendar days prior to commencement of activities.

4. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

5. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

7. The Discharger shall submit the following documents according to the following time schedule to assure compliance with this Order:

| <u>Task</u> | <u>Compliance Date</u> |
|---|---|
| A. Monitoring Well Installation Workplan | **28 February 2001** |

Prior to construction, plans and specifications for installation of a groundwater monitoring wells shall be submitted to Board staff for review and approval. Consideration of potential impacts from future disposal activities on the 15-acre parcel should be given for placement of the wells. Wells are needed to assess background concentrations in shallow groundwater and to assist with the determination of whether there is a potential impact from the wastewater ponds, if any, on the shallow groundwater. The data shall also be used to evaluate potential impacts from other sources. The workplan shall include the following:

- location of the well on a map of the site;
- drilling and development methods;
- well details as shown on a construction diagram;
- schedule of field activities.

B. Well Installation and Groundwater Characterization Report **31 May 2001**

Upon installation of the monitoring wells, a report of the field activities shall be submitted. The report shall include driller's logs and as-builts of well construction. The Report shall also include the analytical results of the initial sampling event from each well. This initial sampling event may be conducted independently or as part of the required monitoring under this Order for the compounds specified in MRP No. 5-01-015. Analytical results shall be presented in tabular form.

The Report shall also include analytical results from the source water assessment of all supply wells, typically conducted at the time of permitting for the California Department of Health Services or the Solano County Department of Environmental Management. If data obtained from this or another monitoring program does not include analytical results for General Minerals (typically including calcium, potassium, sodium, chloride, nitrate, sulfate, total dissolved solids, electrical conductivity, pH, total alkalinity, and total hardness), the Discharger shall collect a sample from each supply well for characterization purposes. Analytical results shall be presented in tabular form.

8. The Discharger shall provide the following documents in the event that the 15-acre parcel is transferred from the Navy to the migrant center, and determination that the parcel, or any portion of the parcel, is to be dedicated to wastewater disposal by use of a pond(s) or land spreading system:

Task Compliance Date

A. Definition of Land Application Area Soil Baseline **3 Months After Land Transfer**

Baseline soil samples from the 15-acre parcel shall be submitted to Board staff. Samples shall be collected and analyzed according to the minimum criteria specified in MRP No. 5-01-015, under Recycling Area/Land Application Area Monitoring.

B. Design Plans and Specifications **6 Months After Land Transfer**

Prior to construction, plans and specifications for either a pond and/or or land application area system shall be submitted to Board staff for review and approval. This document shall provide a description of the wastewater collection, treatment and disposal system, as modified by the addition of the new facility on the 15-acre parcel. Capacity shall be addressed, and if a pond system is constructed, a mass-balance (1-in-100 year return frequency) shall be included. Construction methods and materials shall be described and a schedule for construction shall be provided. Proposals to evaluate the threat to water quality posed by the additional wastewater facility shall also be provided.

C. Operation and Maintenance Plan **9 Months After Land Transfer**

Proposals for operation and maintenance of the additional wastewater facilities with that of the existing system shall be described, including those conditions where routine and or emergency maintenance procedures are needed. The plan shall identify the operator and provide schedules for implementation of the monitoring as required in MRP No. 5-01-015.

9. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

10. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 26 January 2001.

GARY M. CARLTON, Executive Officer

CKW/26 January 2001

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-015
FOR
DIXON MIGRANT CENTER

CITY OF DIXON HOUSING AUTHORITY
YOLO COUNTY HOUSING AUTHORITY
CALIFORNIA DEPARTMENT OF HOUSING AND COMMUNITY DEVELOPMENT,
UNITED STATES DEPARTMENT OF THE NAVY, AND
JOHN AHMANN
SOLANO COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring domestic wastewater, septic tanks, recycling area, and groundwater. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer.

## MONITORING

All samples shall be representative of the volume and nature of the discharge or of the matrix of the material sampled. Time, date, and location of sample collection shall be recorded and presented in the data summary for the monitoring period.

Analyses for the different compounds are to be performed using the following Environmental Protection Agency Methods, Standard Methods, or equivalent methods:

| Compound | Method |
|---|---|
| pH | 150.1 and/or Field Meter |
| Electrical Conductivity (EC) | 120.1 and/or Field Meter |
| Dissolved Oxygen (DO) | 360.1 and/or Field Meter |
| Biochemical Oxygen Demand $(BOD)_{5\text{-day}}$ | 5210B, 405.1 |
| Total Suspended Solids (TSS) | 2540D, 160.2 |
| Nitrates (as Nitrogen) | 300, 352.1 |
| Total Dissolved Solids (TDS) | 2540C, 160.1 |
| General Minerals[1] | as above, and 200.7, 300.0, 2310B, 2340B |
| Total Coliform | 9221E, 909A |
| Fecal Coliform | 9221B, 909C |

1 General Minerals shall include at least the following compounds: calcium, potassium, sodium, chloride, nitrate, sulfate, total dissolved solids, electrical conductivity, pH, total alkalinity, and total hardness.

## SEPTIC TANKS

Unless the septic tank(s) are pumped annually, all septic tanks shall be monitored for the following:

| Parameter | Units | Sample Type | Frequency |
|---|---|---|---|
| Sludge/Scum Thickness[1] | Inches | Measurement | Quarterly |
| Scum to Outlet | Inches | Measurement | Quarterly |
| Sludge to Outlet | Inches | Measurement | Quarterly |

1 Measured in the first compartment of each tank.

## WASTEWATER PONDS

Wastewater ponds are defined as the two ponds currently operated and any additional pond(s) installed on the 15-acre parcel to the north of the existing wastewater ponds. Samples shall be collected from each of the wastewater ponds in operation at the time of sampling for sample analysis. Wastewater pond monitoring shall include the following:

| Constituents | Units | Type of Sample | Frequency[1] |
|---|---|---|---|
| Flow | Gallons/day | Measurement | Cumulative |
| Freeboard | Feet | Measurement | Weekly |
| pH[2] | pH Units | Grab | Monthly[3] |
| EC | µmhos/cm | Grab | Monthly[3] |
| DO[2] | mg/l | Grab | Monthly[3] |
| $BOD_{5\text{-day}}$ | mg/l | Grab | Monthly[3] |
| TSS | mg/l | Grab | Monthly[3] |
| Nitrates (as Nitrogen) | mg/l | Grab | Monthly[3] |
| TDS | mg/l | Grab | Semi-Annually[4] |

1 The monitoring frequency may be reduced upon written request from the Discharger and supported by proven performance data. Reductions in monitoring frequency shall not occur until approved by the Executive Officer.

2 If the compound is outside the permissible limits established in Discharge Specification Nos. B.10 and B.11 of Order No. 5-01-015, the Discharger shall resample within the designated sampling frequency period to determine if compliance was achieved for the monitoring period.

3 The frequency of monitoring shall be reduced to quarterly during the off-season when the migrant center is not fully occupied.

4 Semi-annual monitoring shall be conducted at the height of the wet and dry seasons (typically August/September and January/February, respectively).

## RECYCLING AREA AND/OR LAND APPLICATION AREA MONITORING

### Soil

The Discharger shall monitor the soil within the recycling area and land application area (if constructed) to ascertain whether the nitrate-nitrogen loading is in excess of that required by the vegetation. A minimum of two composite soil samples shall be collected from the recycling area and/or land spreading area. Each composite sample shall be comprised of four to ten discrete soil samples collected throughout the land application area. Samples shall be analyzed for the following compounds:

| Constituents | Units | Type of Sample | Frequency |
|---|---|---|---|
| Salinity | mg/kg | Composite | Annual[1] |
| Nitrogen | mg/kg | Composite | Annual[1] |
| Phosphorus | mg/kg | Composite | Annual[1] |
| Potassium | mg/kg | Composite | Annual[1] |

[1] Samples shall be collected prior to the onset of the rainy season for only those years in which wastewater recycling occurs.

### Storm Water Runoff

The Discharger shall monitor the storm water runoff only from the recycling area on the Ahmann property. The Discharger shall identify the point at which runoff into Reclamation District No.2068's lateral occurs and shall collect a sample at that point. A sample shall also be collected 50 feet upstream of the discharge point in the lateral (provided upstream flow). All samples shall be analyzed for the following:

| Constituents | Units | Type of Sample | Frequency |
|---|---|---|---|
| pH | pH Units | Grab | Two Events[1] |
| EC | µmhos/cm | Grab | Two Events[1] |
| Nitrates (as Nitrogen) | mg/l | Grab | Two Events[1] |
| Total Coliform | MPN/100ml | Grab | Two Events[1] |
| Fecal Coliform | MPN/100ml | Grab | Two Events[1] |

[1] Two sampling events shall be conducted. Baseline sampling shall be conducted during the wet season prior to commencement of recycling, preferably during an initial major storm event. The second sampling event shall be representative of the first flush during the wet season following commencement of recycling.

## GROUNDWATER

Monitoring wells designated for groundwater monitoring of the wastewater ponds shall be installed with the approval of Board staff. Groundwater samples shall be collected from each well for analysis after the well has been adequately purged. Purging is typically defined as either three well volumes, until stabilization of temperature, pH and EC, or refilling of the well after dewatering. The supply wells will be considered adequately purged provided the well is in use prior to the sampling event.

### Monitoring Wells

Each groundwater monitoring well shall be monitored for the following:

| Constituents | Units | Type of Sample | Frequency[1, 2] |
|---|---|---|---|
| Groundwater Elevation[3] | Feet | Measurement | Quarterly |
| pH | pH Units | Grab | Quarterly |
| EC | µmhos/cm | Grab | Quarterly |
| TDS | mg/l | Grab | Quarterly |
| Nitrates (as Nitrogen) | mg/l | Grab | Quarterly |
| Total Coliform | MPN/100ml | Grab | Quarterly |
| General Minerals | mg/l | Grab | Semi-Annually[2, 4] |

1 The Discharger may propose a reduction in frequency to semi-annual provided the data demonstrate no significant changes in water quality. Reductions in monitoring frequency shall not occur until approved by the Executive Officer.

2 Semi-annual monitoring shall be performed at the height of the wet (typically January/February) and dry (typically August/September) seasons. Annual monitoring shall be conducted during the height of the dry season.

3 Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.

4 A "fingerprint" of the groundwater, for characterization purposes, shall be conducted for two years.

### Supply Wells

Monitoring data collected from the supply wells shall be used to assess the character of the source water. Water quality data collected in accordance with Provision 6.B. may also be used to characterize the source water. Samples from each well shall be collected at the closest sampling port to the well head for the following:

| Constituents | Units | Type of Sample | Frequency[1] |
|---|---|---|---|
| pH | pH Units | Grab | Semi-Annually[2] |
| EC | µmhos/cm | Grab | Semi-Annually[2] |

1 The Discharger may propose reduction to annual or may eliminate monitoring after two years, provided the data demonstrate no significant seasonal variations in water quality. Reductions in monitoring frequency shall not occur until approved by the Executive Officer. Monitoring data from other programs may be substituted provided the compounds in that program include those in the table above.

2 Semi-annual monitoring shall be conducted at the height of the wet and dry seasons (typically August/September and January/February, respectively). Annual monitoring shall be conducted at the height of the dry season.

## REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The type of sample (composite or grab) must be identified. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board. All monitoring reports shall include chain-of-custody forms and analytical laboratory data sheets. Time, date, and location of sample collection shall be recorded and presented in the data summary for the monitoring period, consistent with the supporting analytical data sheet.

### A. Quarterly Reports

Quarterly Reports for pond, septic tank, and groundwater monitoring shall be submitted to the Regional Board by the first day of the month following the calendar quarter (i.e, first quarter report is due by **1 April**). During the harvest season (May through October), the quarterly reports shall include monthly water quality results for pond monitoring in addition to the required quarterly groundwater monitoring results. At a minimum, the reports shall include the following:

1. A scaled map showing relevant structures and features of the facility and the locations of all other sampling stations;

2. The results of all wastewater pond monitoring, septic tank monitoring, monitoring well monitoring, domestic wastewater systems monitoring, observations logs and records of sludge, septage, and/or grease removed from the wastewater treatment and disposal facilities;

3. The date(s) and volume (in gpd) of wastewater applied to the recycling area;

4. A groundwater flow map depicting the groundwater elevations and the calculated groundwater flow direction and gradient; and

5. A comparison of the monitoring data to the discharge specifications and an explanation of any violation of these requirements.

**B. Annual Monitoring Report**

An Annual Monitoring Report shall be submitted by **1 February of each year**, and at a minimum, shall include the following:

1. A scaled map showing relevant structures and features of the facility and the locations and any other sampling stations;

2. Analytical results of the soil and runoff samples collected annually from the recycling area and/or land application area;

3. A summary of the all the Monthly Monitoring Reports;

4. Tabular and graphical summaries of all monitoring data obtained during the previous year;

5. Information on the disposal of biosolids;

6. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements; and

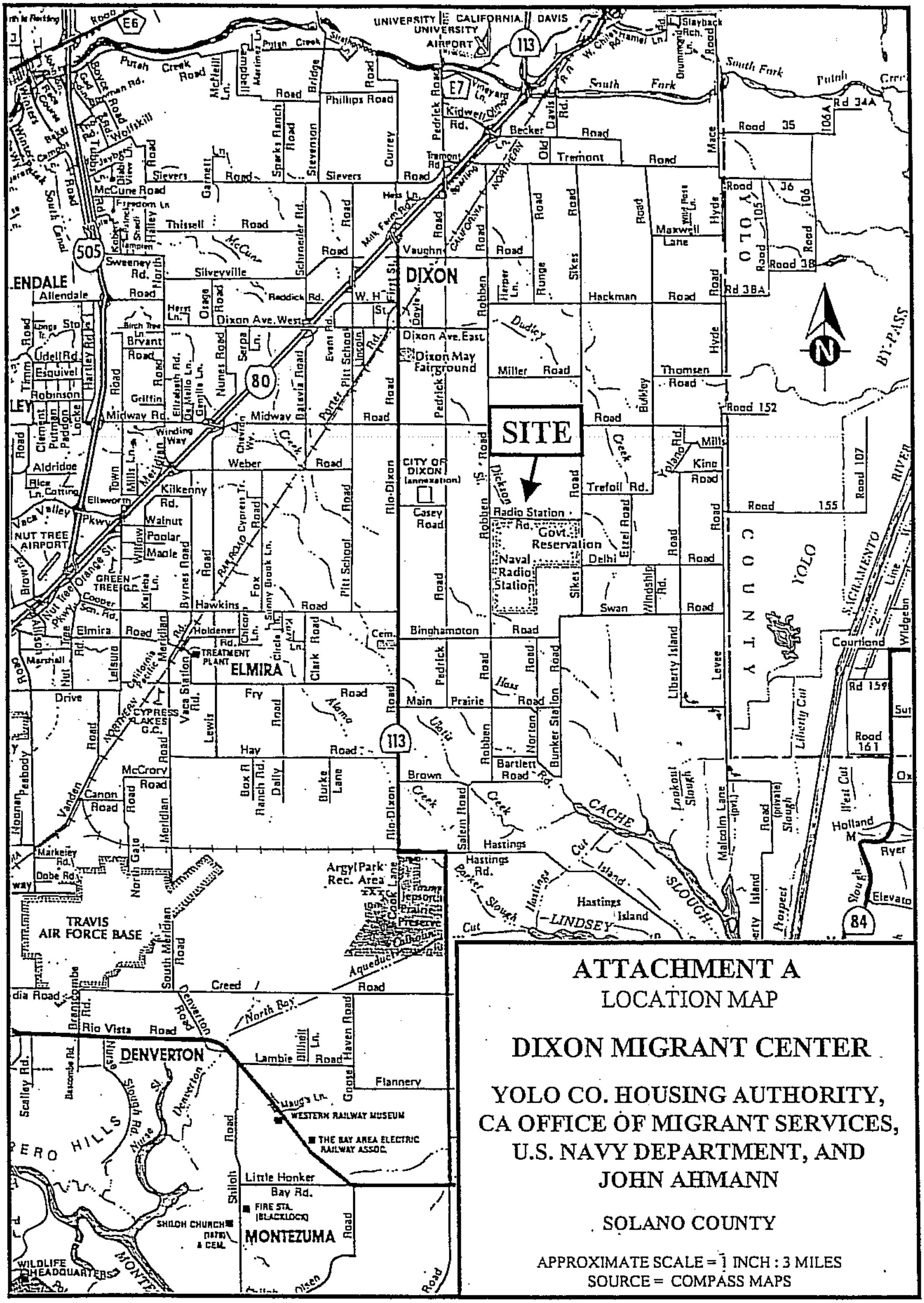
7. A discussion of any data gaps and potential deficiencies, redundancies and/or excesses in the monitoring system or reporting program.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by : GARY M. CARLTON, Executive Officer

26 January 2001
(Date)

CKW




# ATTACHMENT A

LOCATION MAP

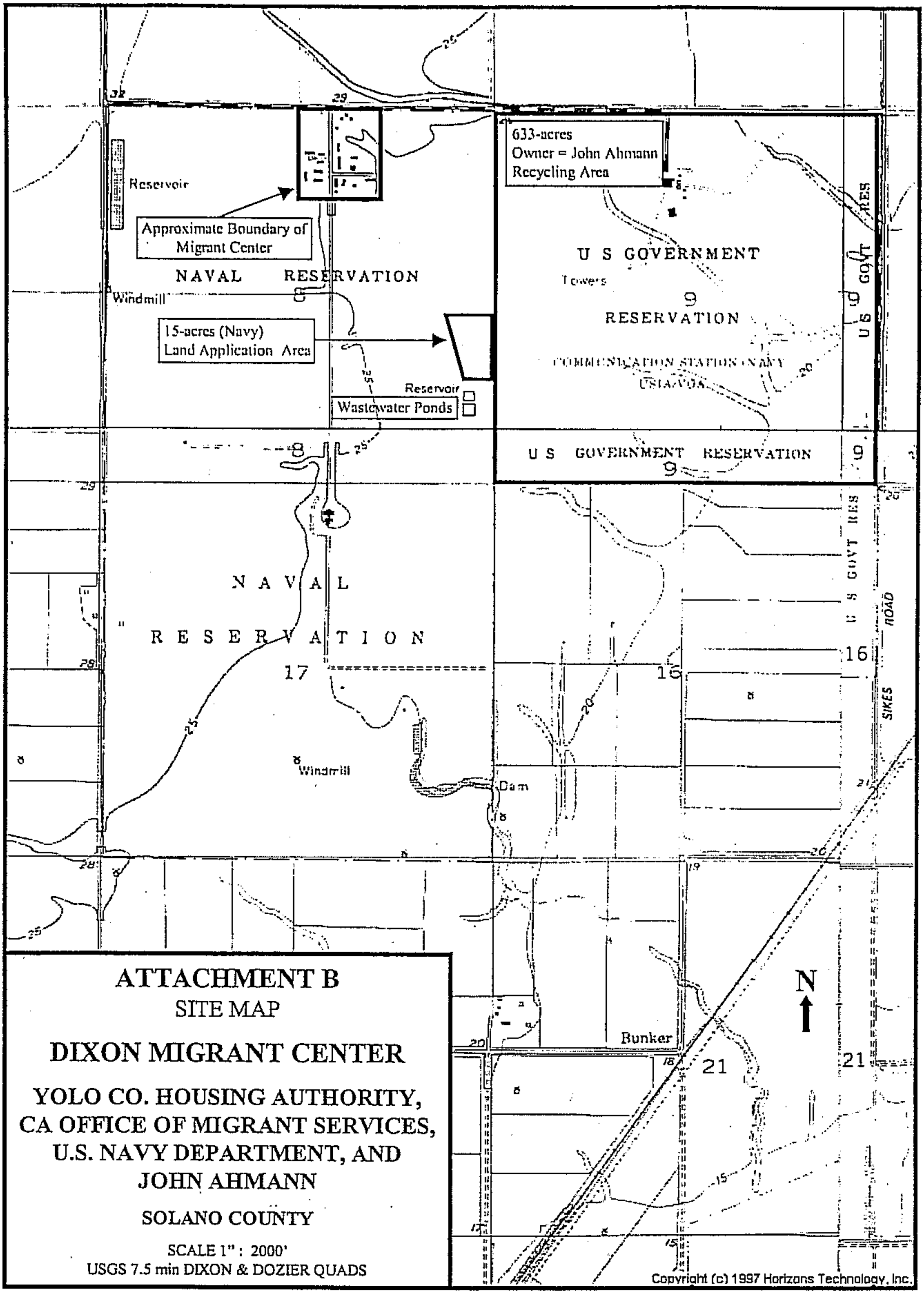
## DIXON MIGRANT CENTER

## YOLO CO. HOUSING AUTHORITY, CA OFFICE OF MIGRANT SERVICES, U.S. NAVY DEPARTMENT, AND JOHN AHMANN

SOLANO COUNTY

APPROXIMATE SCALE = 1 INCH : 3 MILES

SOURCE = COMPASS MAPS




# ATTACHMENT B

SITE MAP

## DIXON MIGRANT CENTER

YOLO CO. HOUSING AUTHORITY,
CA OFFICE OF MIGRANT SERVICES,
U.S. NAVY DEPARTMENT, AND
JOHN AHMANN

SOLANO COUNTY

SCALE 1" : 2000'
USGS 7.5 min DIXON & DOZIER QUADS

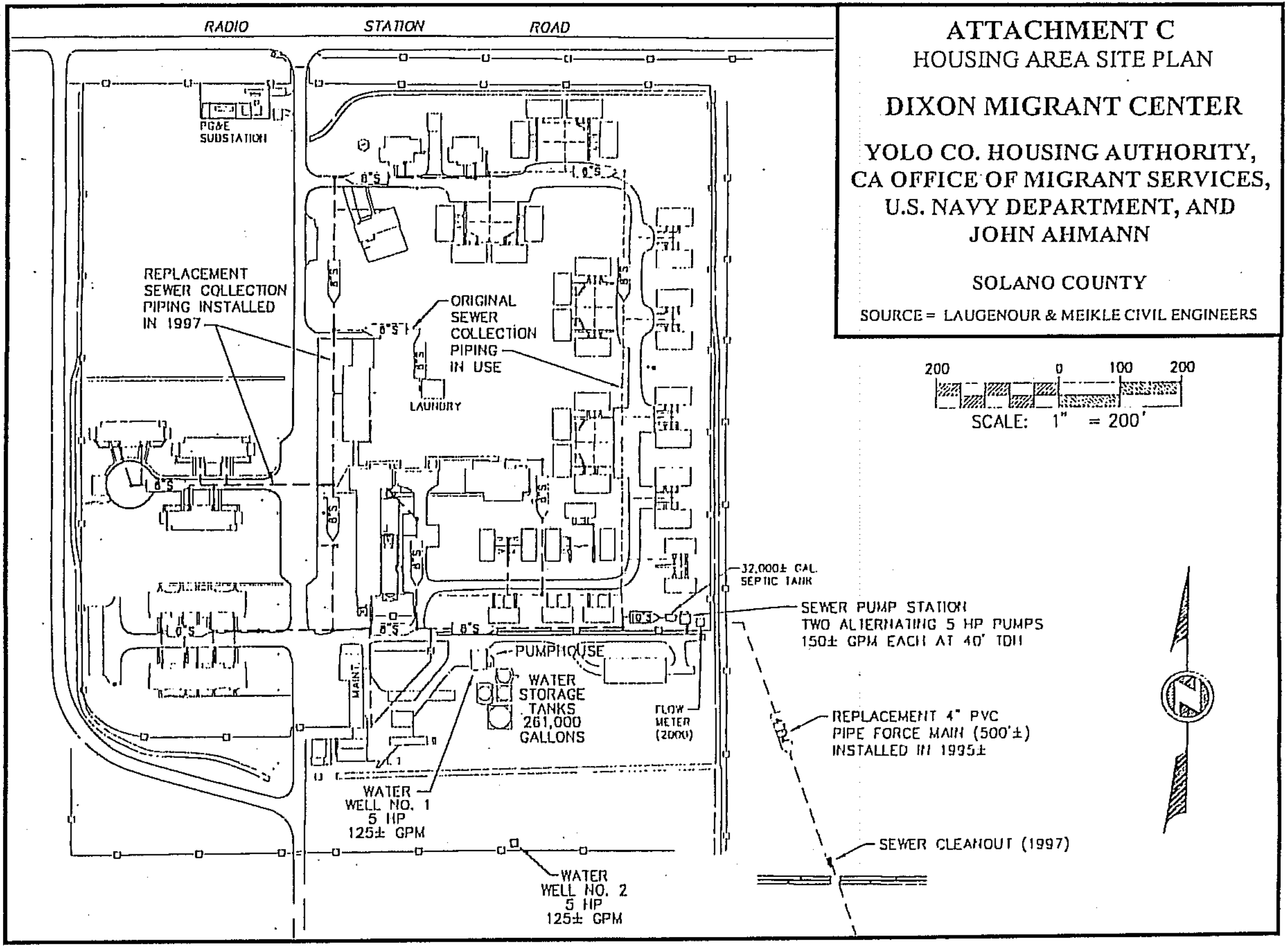
# ATTACHMENT C
## HOUSING AREA SITE PLAN

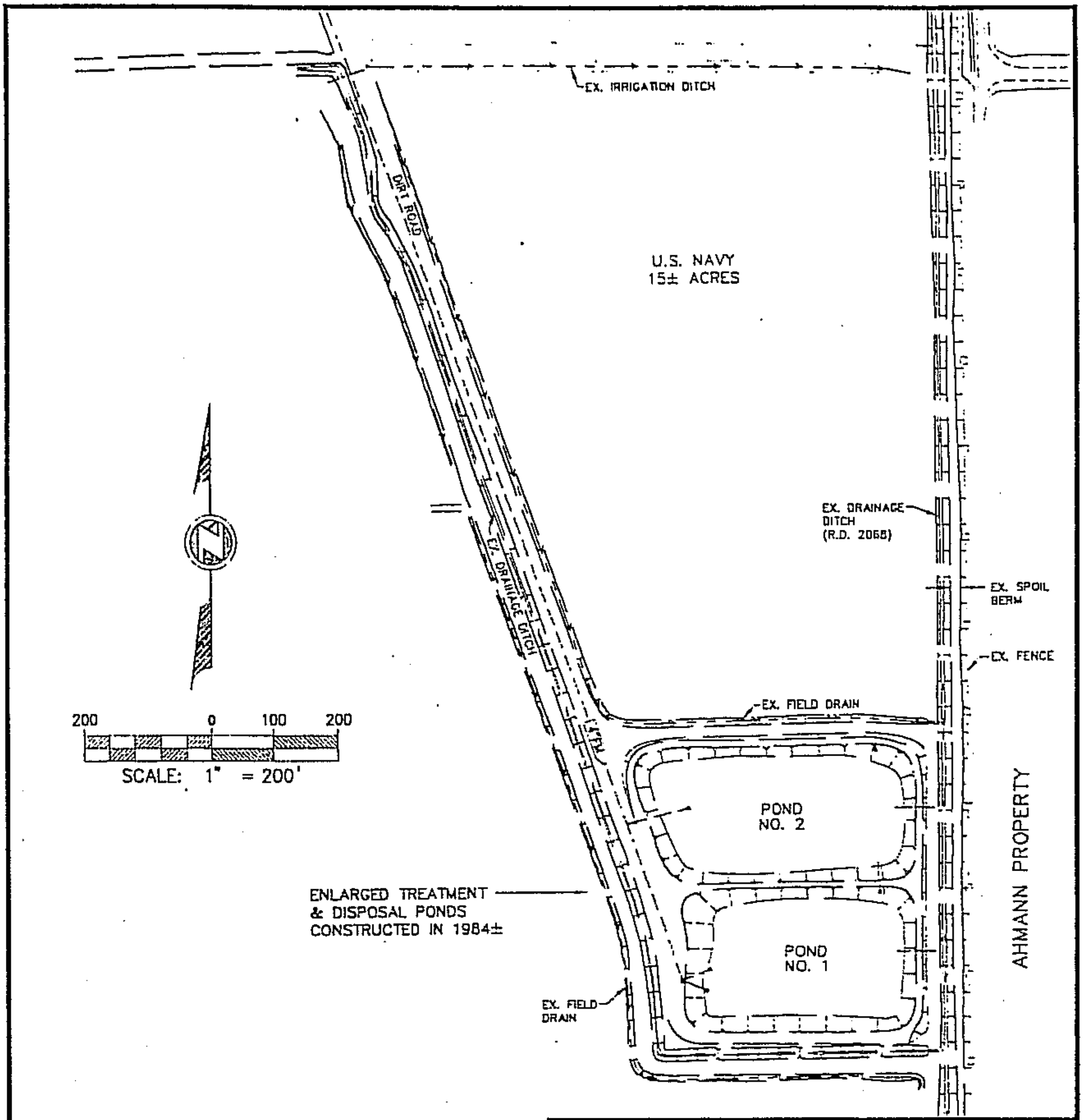
## DIXON MIGRANT CENTER

**YOLO CO. HOUSING AUTHORITY, CA OFFICE OF MIGRANT SERVICES, U.S. NAVY DEPARTMENT, AND JOHN AHMANN**

SOLANO COUNTY

SOURCE = LAUGENOUR & MEIKLE CIVIL ENGINEERS






## EXISTING POND SYSTEM FEATURES

| POND NO | POND BOTTOM ELEV | TOP OF LEVEE ELEV. | MINIMUM FREEBOARD (FEET) | DESIGN HIGH WATER SURFACE ELEV. (1) | DESIGN WATER DEPTH (2) (FEET) | WATER SURFACE AREA AT HWS (ACRES) | POND BOTTOM AREA (ACRES) | USEABLE POND VOLUME (ACRE-FEET) | $BOD_5$ LOADING RATE (3) (LBS/AC/DAY) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 22.0 | 26.55 | 2 | 24.55 | 2.55 | 1.68 | 1.26 | 3.75 | – |
| 2 | 22.3 | 26.55 | 2 | 24.55 | 2.25 | 1.85 | 1.46 | 3.72 | – |
| TOTAL | – | – | – | – | – | 3.53 | 2.72 | 7.47 | 19 |

(1) THIS ELEVATION PROVIDES FOR 2' OF FREEBOARD (ORIGINAL 1984 DESIGN SPECIFIED 1' OF FREEBOARD.)

(2) THIS DEPTH CORRESPONDS TO A STAFF GAUGE READING OF 1'-10". ADD 22.75' TO STAFF GAUGE READING TO OBTAIN WATER SURFACE (WS) ELEVATION

(3) ESTIMATED LOADING RATE DURING THE 6-MONTH MIGRANT WORKER SEASON, BASED ON 0.17 LBS/PERSON/DAY RATE AND NO REMOVAL BY 37,000 GALLON SEPTIC TANK.

# ATTACHMENT D

## POND PLAN & SPECIFICATIONS

## DIXON MIGRANT CENTER

### YOLO CO. HOUSING AUTHORITY, CA OFFICE OF MIGRANT SERVICES, U.S. NAVY DEPARTMENT, AND JOHN AHMANN

SOLANO COUNTY

SOURCE = LAUGENOUR & MEIKLE CIVIL ENGINEERS

## INFORMATION SHEET

DIXON MIGRANT CENTER, ORDER NO. 5-01-015
CITY OF DIXON HOUSING AUTHORITY, YOLO COUNTY HOUSING AUTHORITY,
CALIFORNIA DEPARTMENT OF HOUSING AND COMMUNITY DEVELOPMENT,
UNITED STATES DEPARTMENT OF THE NAVY, AND
JOHN AHMANN
SOLANO COUNTY

The Dixon Migrant Center is operated by the Yolo County Housing Authority under a yearly contract issued by the California Department of Housing and Community Development (Department), through the Dixon Housing Authority. The migrant center is on a 27-acre parcel of a larger, 161-acre parcel at the U.S. Naval Radio Transmitting Facility (NRTF), Dixon, approximately 6 miles southeast of Dixon in Solano County. The United States Department of the Navy owns the migrant center property, but is in the process of declaring the property surplus through the U.S. General Services Agency.

In November 1999, the Yolo County Housing Authority submitted a Report of Waste Discharge (RWD) to include alternative discharge options for disposal of the domestic wastewater. An amendment to the RWD was submitted in June 2000. Two alternative discharge options were considered in the RWD to provide additional capacity during maintenance activities or emergency situations. These alternatives included the off-site recycling of the wastewater to the 633-acre parcel immediately east of the ponds, owned by John Ahmann, and discharge to a 15-acre parcel north of the ponds, currently owned by the Navy, via land application or construction of a pond. The Ahmann property, where wastewater recycling is proposed, is bounded on the east and south by lateral drainage ditches maintained by Reclamation District No. 2068. A land lease agreement between the Yolo County Housing Authority and John Ahmann has been executed to allow recycling of the wastewater. The 15-acre parcel is also considered surplus and may be transferred with the migrant center and wastewater pond parcels.

The Department has made many improvements to the collection and conveyance system since 1999. These improvements include repair of the septic tank, sewer manholes, and distribution boxes, and replacement of collection piping. A flow meter was installed in January 2000. These improvements and the proposed alternative discharge options necessitates that Waste Discharge Requirements (WDRs) Order No. 95-128 be revised.

The migrant center houses approximately 370 residents during the harvest season, which extends from 1 May through 31 October. Only about 15 people reside at the center in the non-harvest season. During the harvest season, maximum wastewater flows are estimated to be approximately 37,000 gallons per day (gpd) and average flows are estimated to be 25,000 gpd. In the non-harvest season, flows are estimated to be up to 6,000 gpd. Wastewater is discharged to a 32,000-gallon septic tank, which is pumped to two oxidation ponds.

Recent flow meter data indicates that up to 5,000 gpd of groundwater may infiltrate into the ponds during the non-harvest season. The ponds have experienced problems with insufficient storage capacity in the past, as the wastewater was observed to discharged directly, or threaten to discharge, to an off-site agricultural ditch during inspections by the Regional Board and Navy. The by-pass valve to the ditch was removed in September 1999, as part of the facility improvements.

The water supply for the site is from two wells at the NRTF, Dixon. A new supply well will be installed upon transfer of the property from the Navy. There are no groundwater monitoring wells at the site, but depth to shallow groundwater was estimated in the RWD as approximately 4 feet below ground surface. The shallow soil is clayey.

Under these WDRs, monitoring of the septic tank, pond wastewater, groundwater, soil in the recycling and/or land application areas, and surface water runoff from the recycling area are required. If the 15-acres currently owned by the Navy are transferred to the migrant center and are used for the discharge of wastewater, these WDRs will also require submittal of design and operation plans, and monitoring of those facilities. In addition, the installation of monitoring wells and groundwater characterization are required under these WDRs. Monitoring data will be used to determine whether wastewater treatment and disposal has the potential to impact groundwater quality.

The beneficial uses of the Haas Slough, Cache Slough, and the Sacramento River are municipal and domestic, agricultural supply for irrigation and stock watering, contact and non-contact recreation, warm freshwater habitat, fish spawning, wildlife habitat, and groundwater recharge.

The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.

Surface water drainage from the site is to an agricultural ditch, which drains to a lateral operated by Reclamation District No. 2068, which drains to Hass Slough, tributary to Cache Slough, tributary to the Sacramento River.